REACHING A BOUNDARY POINT OF THE IMAGE WITH A DVR OVER A NOETHERIAN BASE

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Editorial note. Counted span is the author's own proof environment, source lines 4792-4811, following the statement at 4771-4790.

Editorial note. No omissions; the codimension-0 and residue-field refinements are part of the scheme-engine output, per the footnote reference.

Allowed standard prerequisites (editorial).

- Points of cartesian squares under surjective etale maps. Properties of Spaces, Lemma spaces-properties-lemma-points-cartesian (cross-chapter, cited at ~4794-4797). The image of |X x_Y V| -> |V| is the preimage of the image of |f|.
- Scheme-level dvr reach lemma. Limits, Lemma limits-lemma-reach-point-closure-Noetherian (cross-chapter, cited at ~4809-4810). The engine for the affine scheme case.

Source reference. Original reference 0CMC: focused-seeds/geometry/S9-spaces-limits-reach-point-closure-Noetherian/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H4: focused-seeds/geometry/S9-spaces-limits-reach-point-closure-Noetherian/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BAP: focused-seeds/geometry/S9-spaces-limits-reach-point-closure-Noetherian/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f finite type and Y locally Noetherian. Let $y \in |Y|$ be a point in the closure of the image of $|f|$. Then there exists a commutative diagram*

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & & \downarrow f \\ \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

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where A is a discrete valuation ring and K is its field of fractions mapping the closed point of $\mathrm{Spec}(A)$ *to y. Moreover, we can assume that the point $x \in |X|$ corresponding to* $\mathrm{Spec}(K) \to X$ *is a codimension* 0 *point*[1] *and that K is the residue field of a point on a scheme étale over X.*

Proof. Choose an affine scheme V, a point $v \in V$ and an étale morphism $V \to Y$ mapping v to y. The map $|V| \to |Y|$ is open and by Properties of Spaces, Lemma 03H4 the image of $|X \times_Y V| \to |V|$ is the inverse image of the image of $|f|$. We conclude that the point v is in the closure of the image of $|X \times_Y V| \to |V|$. If we prove the lemma for $X \times_Y V \to V$ and the point v, then the lemma follows for f and y. In this way we reduce to the situation described in the next paragraph.

Assume we have $f : X \to Y$ and $y \in |Y|$ as in the lemma where Y is an affine scheme. Since f is quasi-compact, we conclude that X is quasi-compact. Hence we can choose an affine scheme W and a surjective étale morphism $W \to X$. Then the image of $|f|$ is the same as the image of $W \to Y$. In this way we reduce to the case of schemes which is Limits, Lemma 0CMC. □

[1] See discussion in Properties of Spaces, Section 0BAP.